

1 HYDRA-BDE Input Types (relating to personnel)

1.1 1.1 Summary

The sections that follow describe the different personnel-related views by way of examples. Postings based on people and on operations are displayed. To simplify matters, breaks are not included in the calculation of time intervals.

1.2 Single machine operation

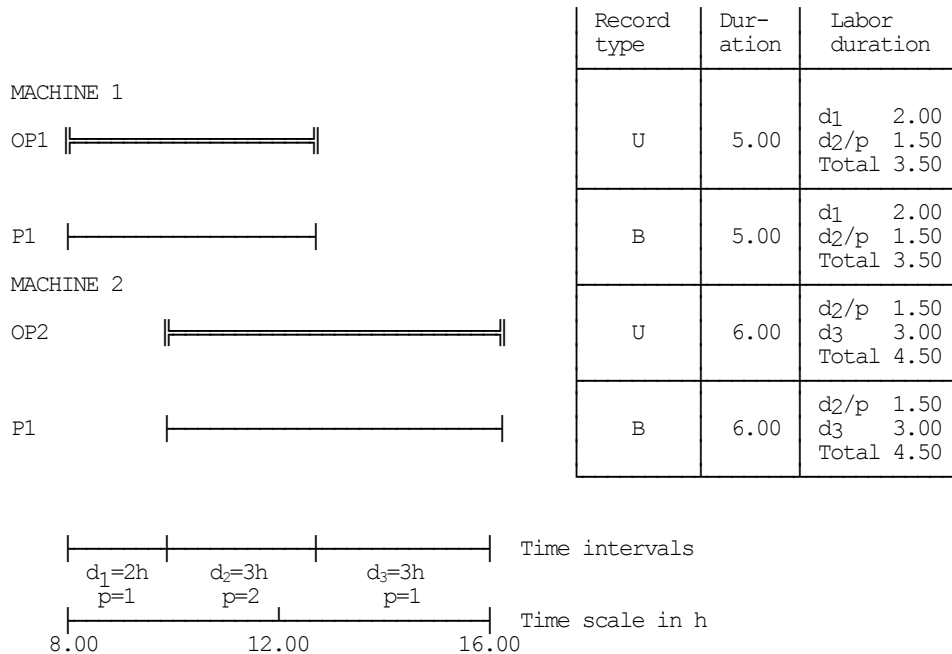
"Single machine operation" means that one person works alone on an operation at one workstation.

1.3 Multiple machine operation

Multi machine operation means that one person processes several operations at several workstations at the same time. This is normal, for example, on semi or fully automatic machines, where one person monitors several machines, feeding them with raw material and removing the finished parts. The labor utilization relating to operations is reduced according to the number of machines that are operated simultaneously.

Example

One person processes two operations at the same time on two machines.



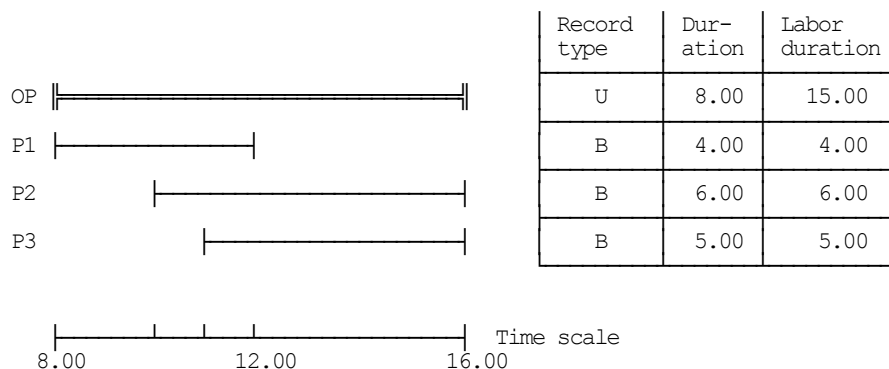
p : the number of times the same person has logged in
d_i: time interval i

1.4 Group work

Group work means that several people process a single operation at one workstation. With every additional person who is logged on, the labor utilization increases accordingly.

Example

Three people process an operation on one machine.



1.5 "Mixed operation"

HYDRA allows for all of the above-mentioned variants to be mixed, that is, HYDRA correctly handles the situation where a person with multi-machine operation is additionally logged on to a machine with group work.

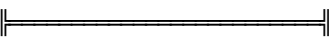
Example

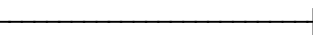
Several people process several operations on several machines (parallel production).

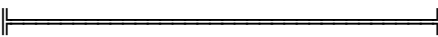
Machine 1 is an individual workplace

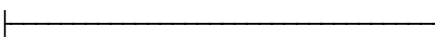
Machine 2 is a group workplace

Machine 1 Individual Workplace

OP 1 

P1 

OP2 

P1 

Record type	Dur- ation	Labor duration
U	d1 2	d1 2.0 P1
	d2 1	d2/2 0.5 P1
	d3 2	d3/3 0.7 P1
	Total 5	Total 3.2
B	d1 2	d1 2.0
	d2 1	d2/2 0.5
	d3 2	d3/3 0.7
	Total 5	Total 3.2
U	d2 1	d2/2 0.5 P1
	d3 2	d3/3 0.7 P1
	d4 1	d4/2 0.5 P1
	d5 2	d5/2 1.0 P1
	d6 1	d6 1.0 P1
	Total 7	Total 3.7
B	d2 1	d2/2 0.5
	d3 2	d3/3 0.7
	d4 1	d4/2 0.5
	d5 2	d5/2 1.0
	d6 1	d6 1.0
	Total 7	Total 3.7

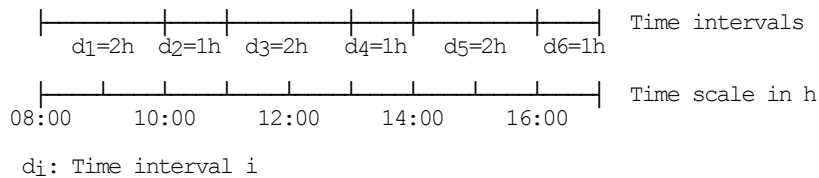
Machine 2 Group Workplace

OP 3

P2

OP 4

P1



U	d1	2	d1	2.0	P2
	d2	1	d2	1.0	P2
	d3	2	d3	2.0	P2
	d4	1	d4	1.0	P2
	Total	6	Total	6.0	
B	d1	2	d1	2.0	
	d2	1	d2	1.0	
	d3	2	d3	2.0	
	d4	1	d4	1.0	
	Total	6	Total	6.0	
U	d3	2	d3/3	0.7	P1
	d4	1	d4/2	0.5	P1
	d5	2	d5/2	1.0	P1
	Total	5	Total	2.2	
B	d3	2	d3/3	0.7	
	d4	1	d4/2	0.5	
	d5	2	d5/2	1.0	
	Total	5	Total	2.2	